Reg. No. | | | | | | | | | | | [18MTSE202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

FINITE ELEMENT METHOD

Time: 03 Hours **Maximum Marks: 100**

Instructions: **i. Answer FIVE full questions. Question 1&2 are compulsory.**
ii. Missing data, if any, may be suitably assumed.

1. a. With the help of an example explain the basic steps involved in the Finite Element Method. Give the applications, advantages and disadvantages of the method. Explain the application of Rayleigh Ritz method in the Finite Element formulation. **20**

2. a. Explain the convergent and compatibility requirements. **10**

b. What is meant by one – dimensional finite element, obtain the stiffness matrix for one – dimensional element? **10**

3. a. Explain the terms isoparametric, sub-parametric and super parametric elements. **10**

b. Write short notes on Jacobian matrix and Gaussian quadrature integration technique. **10**

OR

4. For a four noded rectangular element shown in figure 1, determine the following at $\xi=0$; $\eta=0$. Take $E=2X10^5$ N/m^2. υ = 0.25 and u = [0,0, 0.002, 0.003, 0.005, 0.003, 0, 0] Assume plane stress condition. Find (a) Jacobian matrix (b) Strain-displacement matrix (c) Element stress. **20**

Node	X	Y
1	0	0
2	2	0
3	2	1
4	0	1

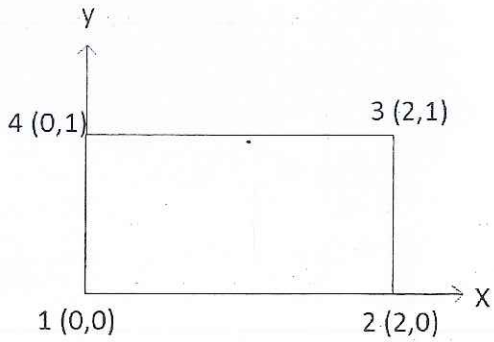
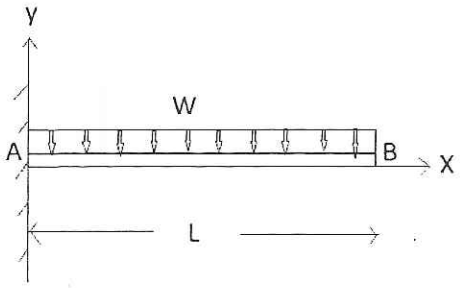


Figure 1

5. For the cantilever beam subjected to the uniform load W as shown in figure 2. Solve for the right-end vertical displacement and rotation for the nodal forces. Assume the beam to have constant EI throughout its length. **20**



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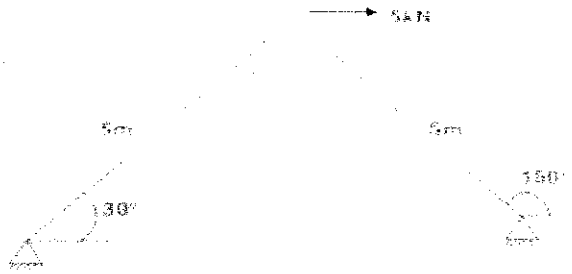
OR

6. Analyse the two member truss shown in figure 3. Assume EA to be constant for all the members. The length of each member is 5m. **20**



7. What are the various categories of non-linear problems? How do you solve non-linear problems? Explain any one of the procedures for two categories. **20**

OR

8. Derive the basic relation for thin plate theory using typical plate elements. **20**